

ZODIACCHAIN

Strategic Whitepaper Polygon-First Testnet Strategy & Grant Documentation

1. Executive Summary

ZodiacChain is a decentralized, verifiable betting infrastructure built on Polygon that combines 25 Terrestrial Animals with 12 Celestial Animals into a highly auditable gaming experience. Positioned for its initial phase as a testnet-first, infrastructure-focused MVP, ZodiacChain is currently designed for verifiable public demonstration and ecosystem grants rather than immediate real-money operations. The protocol utilizes a mathematically consistent system where the core official result is an Exact Celestial Number (01–60), providing complete transparency and verifiable randomness.

ZodiacChain is not positioned merely as an end-user destination, but as a verifiable execution layer that demonstrates how onchain randomness can be orchestrated transparently and autonomously.

ZodiacChain is designed as infrastructure, not as a traditional gambling platform.

2. The Problem & ZodiacChain Solution

Traditional lottery markets suffer from low operational transparency and un-auditable player returns. Conversely, crypto gaming platforms often feature generic user experiences with low cultural identity and obscure smart contract mechanics. ZodiacChain bridges this gap by merging cultural familiarity with straightforward mathematical readability and verifiable blockchain infrastructure. By focusing its current narrative on testnet onboarding, verifiable fairness, and open-source documentation, ZodiacChain minimizes regulatory friction and maximizes alignment with major ecosystem grant programs.

3. Product Architecture

The modular Web3 stack is optimized for cost, speed, and verifiability:

- **Network:** Polygon PoS serves as the priority network for the MVP due to its EVM tooling, low transaction costs, and strong ecosystem grant alignment.
- **Randomness Engine:** Chainlink VRF powers the draw mechanism, ensuring the execution flow is clear and publicly verifiable.
- **Currency:** All wagers utilize stablecoins within the Polygon environment to eliminate crypto volatility and reduce user friction.
- **Fairness Dashboard:** A dedicated interface displays the draw history, the exact Celestial Number, the derivation of animals and elements, and real-time pool protections.

The architecture is designed to prioritize verifiable data flow, deterministic outcome derivation, and transparent execution. In future phases, this model may serve as a reusable foundation for broader decentralized gaming modules.

4. Game Mechanics

ZodiacChain utilizes a distinctive two-dimension betting system, offering varied risk-reward profiles.

The system operates through two independent dimensions:

Terrestrial Dimension (Level 1):

The Terrestrial Dimension is based on a 00–99 numerical system, where each group of four consecutive numbers maps to one of the 25 animals.

This creates a deterministic and fully auditable structure:

- 100 total numbers
- 25 animals
- 4 numbers per animal

Example:

00–03 → Animal 1
04–07 → Animal 2
...
96–99 → Animal 25

Players can place bets on a specific animal (22.5x multiplier), exact tens (90x), hundreds (900x), or thousands (up to 9,000x multiplier).

The full animal-to-number mapping is defined at the protocol level and made publicly accessible through the application interface and supporting documentation.

Celestial Dimension (Level 2):

Inspired by the Chinese sexagenary cycle, this dimension is anchored by a single official result: the Exact Celestial Number (01–60).

The Celestial Number (01–60) is the single source of truth for all Level 2 outcomes.

From this official result, the protocol deterministically derives the Celestial Animal and Element according to ZodiacChain's product mapping. This approach improves clarity, removes ambiguity, and ensures that every Celestial outcome can be publicly verified from a single official result.

All derived outcomes are deterministic and publicly verifiable from this single value.

Celestial Number Derivation (01–60)

- Animal mapping:
 - Rat: 01, 13, 25, 37, 49
 - Ox: 02, 14, 26, 38, 50
 - Tiger: 03, 15, 27, 39, 51
 - Rabbit: 04, 16, 28, 40, 52
 - Dragon: 05, 17, 29, 41, 53
 - Snake: 06, 18, 30, 42, 54
 - Horse: 07, 19, 31, 43, 55
 - Goat: 08, 20, 32, 44, 56
 - Monkey: 09, 21, 33, 45, 57
 - Rooster: 10, 22, 34, 46, 58
 - Dog: 11, 23, 35, 47, 59
 - Pig: 12, 24, 36, 48, 60

- Element mapping:
 - 01–12 = Wood
 - 13–24 = Fire
 - 25–36 = Earth
 - 37–48 = Metal
 - 49–60 = Water

Example:

If the official result is 17, the derived Celestial Animal is Dragon and the derived Element is Fire.

Players can bet on the derived Element (4.5x), Celestial Animal (10.8x), or the Exact Celestial Number (54x).

Combined Bets & Special Events (Level 3)

- Combo Bets: Cross-dimensional wagers yield massive multipliers, such as combining an exact Terrestrial Ten with the Exact Celestial Number for a 5,400x payout.
- Cosmic Encounter: If the same animal is drawn in both dimensions on the same day, winners receive an additional bonus from a dedicated fund.
- Weekly Jackpot: A special rollover pool for players who successfully match 1 Terrestrial Animal + 1 Celestial Animal + 1 Celestial Element.

5. Economic Design & Risk Controls

ZodiacChain features a robust economic model designed to ensure pool solvency and high predictability for grant documentation.

Capital Allocation:

- 90% Return to Player (RTP) per draw.
- 5% Operational Treasury.
- 3% ZDC Liquidity Fund.
- 2% Cosmic Encounter Fund.

Solvency & Risk Protections:

- Betting Limits: \$1 minimum (to prevent spam/gas distortion), \$500 maximum per bet, and a \$2,000 daily limit per wallet for responsible gaming and AML compliance.
- Payout Controls: Single payouts are hard-capped at \$50,000 to preserve initial pool liquidity.
- Concentration Rule: No single animal can hold more than 40% of the draw's total pool, severely mitigating malicious hedging.
- Anti-Bot measures: A strict 30-second cooldown interval between bets prevents frontrunning.
- Solvency Verification: The smart contract automatically blocks wagers if the available pool cannot cover the maximum potential payout.

6. Roadmap & Grants Strategy

- Phase 1 (0-30 Days): Documentation finalization, grant application packaging, basic identity, and a simple website.
- Phase 2 (30-90 Days): MVP testnet deployment, public verifiable draw demo, and fairness dashboard launch.
- Phase 3 (90-150 Days): Institutional preparation, legal reviews, developer onboarding, and regulatory mapping.
- Phase 4 (150+ Days): Subsequent funding rounds, product scaling, and evaluation of regulated launch paths.

Grants Focus:

1. Chainlink Build: Priority 1, due to direct alignment with VRF and verifiable infrastructure.
2. Polygon Ecosystem: Priority 2, focusing on MVP testnet deployment and open-source documentation.
3. Gitcoin: Priority 3, strictly for independent open-source subprojects.

No real-money operations are planned during the initial phases. The protocol is currently focused on testnet validation, public technical demonstration, and infrastructure development.

The public testnet MVP will serve as a practical validation of this randomness infrastructure, demonstrating how consumer-facing applications can reduce black-box trust assumptions through verifiable onchain execution and deterministic outcome derivation.

7. Initial Operating Strategy

ZodiacChain operates with a highly prudent regulatory stance, operating currently only as a testnet demonstrator.

- **Favorable-Jurisdiction-First:** Future real-money operations will be exclusively structured in compliant jurisdictions (e.g., Curacao, Malta, Gibraltar, or Isle of Man) subject to proper legal counsel and local licensing.
- **Brazil Excluded from Phase 1:** Despite the core team's location, Brazil is strictly excluded from Phase 1 commercial launches unless a specific local authorization framework from the SPA/MF is achieved.
- **Strict Access Controls:** Any future token-related activity or regulated commercial deployment is outside the current phase and would require strict geofencing, KYC/AML policies, full legal review, and explicit regulatory authorization before consideration.

8. Core Team

The Core Team is based full-time in Recife, Brazil, bringing deep technical and operational expertise.

- **Edson Marinho Gomes — Founder, Product & Backend Lead (6 years).**
 - Product strategy, backend architecture, and cross-functional leadership.
 - [LinkedIn](#)
- **Bismark Jamerson — Backend Engineer & Software Analyst (6 years).**
 - Backend systems design, requirements analysis, and integration execution.
 - [LinkedIn](#)
- **José Rodrigues Lins Neto — QA Lead (13+ years).**
 - Test strategy, defect management, and release quality control.
 - [LinkedIn](#)
- **Italo Melo — Business Development & Partnerships Lead (4+ years, fluent English).**
 - Partnership outreach, stakeholder communication, and international positioning.

9. Open Roles

To complete the protocol's architecture and regulatory compliance, the project is actively seeking:

- Smart Contract / Solidity Lead.
- Frontend Web3 Engineer.
- Legal & Regulatory Counsel.
- External Security Auditor.

10. Contact & Public Presence

- Email: zodiacchain.protocol@gmail.com
- X/Twitter: @ZodiacChainXYZ
- GitHub: ZodiacChain

11. Disclaimer

Important: This document is purely informational and describes a protocol currently in its technical documentation and validation phase, adjusted specifically for ecosystem grants and testnet demonstration. It does not constitute an offer of investment, a promise of financial return, or an invitation to participate in real-money operations in any jurisdiction. Any future phases involving real money will be strictly dependent upon acquiring the appropriate regulatory frameworks, comprehensive legal reviews, and stringent jurisdictional access controls.